# Validation Report - Network Security Project (Phase 1)

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### 1. Topology Diagram and Addressing Summary

The network consists of three sites (Left, Core, Right) connected via two WAN links.

| Site | LAN Network | Gateway (Router Interface) | Server IP | DHCP Range |
|---|---|---|---|---|
| **Left** | 192.168.1.0/24 | R0 (192.168.1.1) | Server0 (192.168.1.10) | 192.168.1.21 -254 |
| **Core** | 172.16.1.0/16 | R4 (172.16.1.1) | N/A | Static Ips |
| **Right** | 192.168.2.0/24 | R2 (192.168.2.1) | Server1 (192.168.2.10) | 192.168.2.21 - 254 |

**WAN Links:**

- R0 (10.0.0.1/30) <-> R4 (10.0.0.2/30)
- R4 (10.0.0.5/30) <-> R2 (10.0.0.6/30)

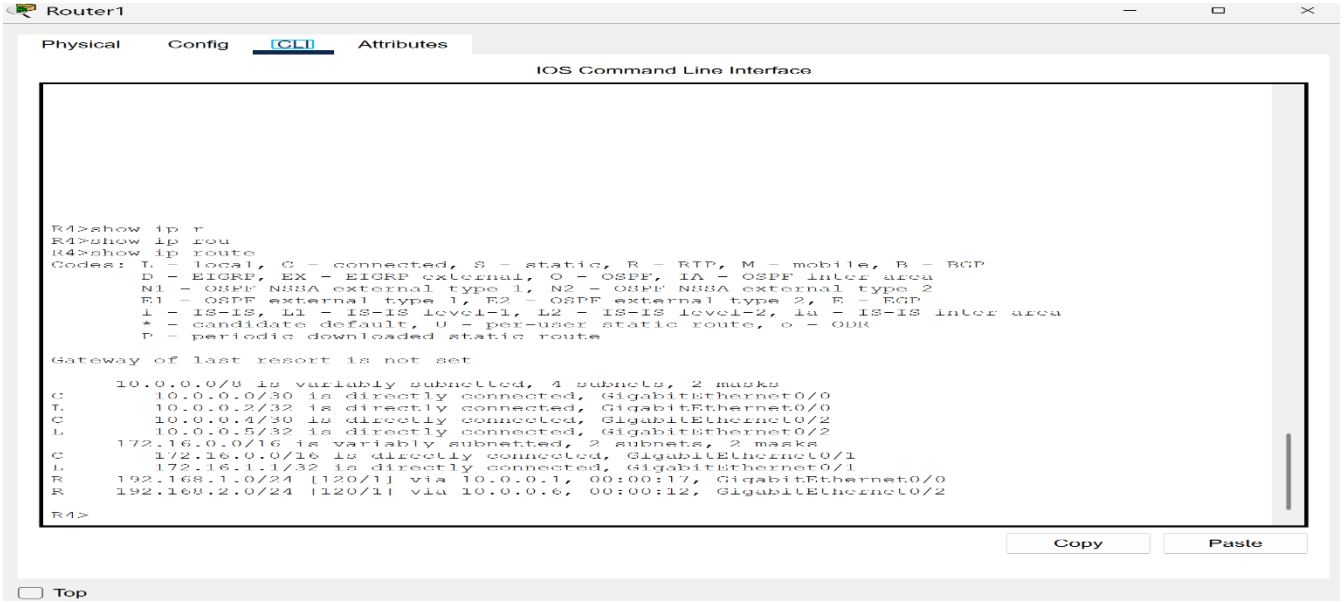
### 2. Phase-1 Validation Screenshots

**A. RIP Routes Visible ( `show ip route` on all routers)**

**Expected Result:** The routing table on each router must show routes learned via RIP (indicated by R) for all remote LAN and WAN networks.

- **R0 Expected Output:** Should show routes to 172.16.0.0/16 and 192.168.2.0/24.
- **R4 Expected Output:** Should show routes to 192.168.1.0/24 and 192.168.2.0/24.

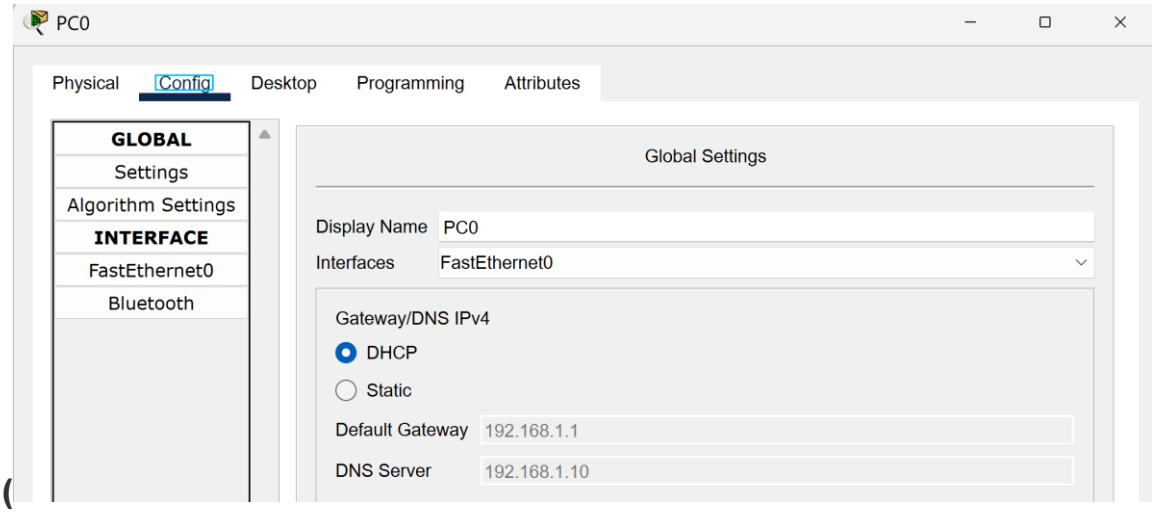
- **R2 Expected Output:** Should show routes to 192.168.1.O/24 and 172.16.O.O/16.




### B. PCs Get Correct IPs via DHCP (PC0, PC3, PC4)

**Expected Result:** PCO, PC3, and PC4 must successfully obtain an IP address, Subnet Mask, Default Gateway, and DNS Server address from the configured DHCP pools.

- **PC0 Expected IP:** An address in the range 192.168.1.21 - 192.168.1.254.
- **PC3/PC4 Expected IP:** An address in the range 192.168.2.21 - 192.168.2.254.




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### C. `nslookup srv1.right.local` from the left side resolves correctly

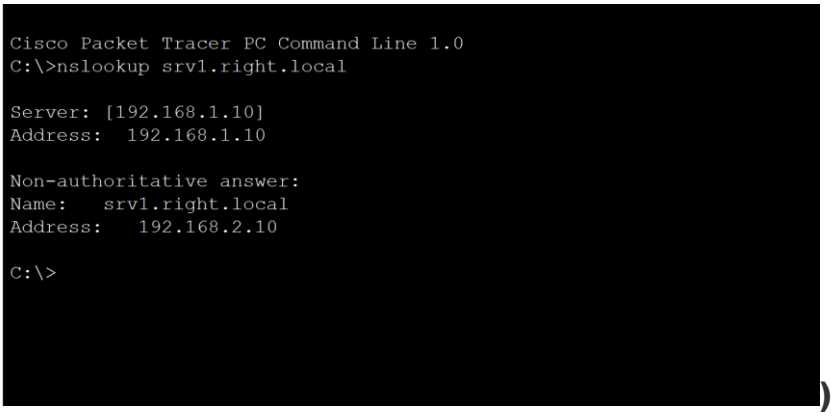
**Expected Result:** Performing an from PC0 (Left Site) for the server in the nslookup

Right Site ( `srv1.right.local` ) must resolve to the correct IP address (192.168.2.1O),

proving cross-site DNS resolution.

```
PC0> nslookup
srv1.right.local Server:
192.168.1.10
Address: 192.168.1.10#53

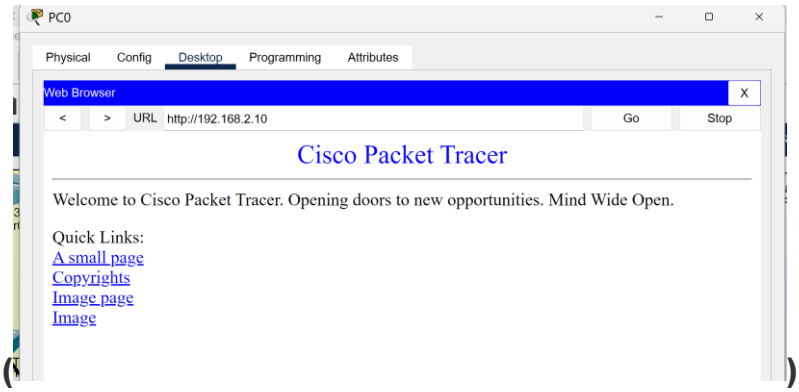
Name:
srv1.right.local
Address: 192.168.2.10
```

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**D. Browse `http://192.168.2.10` from PC0 successfully**

**Expected Result:** PCO must be able to open the web browser and successfully load the default HTTP page from Server1 (192.168.2.1O), proving end-to-end connectivity and service reachability.

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## 3. Assumptions and Limitations

The following assumptions were made during the configuration of Phase 1:

1. **DCE/DTE Assignment:** It was assumed that RO and R4 (on the R4-R2 link) are the DCE ends for the Serial links, and the `clock rate 128000` command was applied accordingly. If the physical connection is different, the `clock rate` command must be moved to the correct router interface.
2. **Device Naming:** The configuration adheres strictly to the required device names (RO, R4, R2, ServerO, Server1) as per the project document, even if the provided topology image used slightly different names/placements for the servers.
3. **Switch Configuration:** Switches (SO, S1, S2) were assumed to be operating at Layer 2 with default settings, as no specific configuration was mandated for Phase 1.
4. **Interface Mapping:** The configuration uses common interface names (GiO/O/O, SeO/1/O, etc.). The student must ensure these match the actual physical interfaces used in their Packet Tracer file.